

A/CR

Andres
Cervera Rozo

ROBOTICS ENGINEER
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SKILLS

Computer Vision

Robotics

Python

Hardware Prototyping

ML & RL

ROS/ROS2

Embedded Systems

C/C++

GCP/Azure

MATLAB/Simulink

EDUCATION

Master of Engineering, Emphasis in Robotics

University of Toronto 2023-2024

BASc, Mechanical Engineering (High Honours)

University of Toronto 2018-2023

Robotics & Mechatronics Minor

University of Toronto 2018-2023

Certified Solidworks Professional (CSWP)

EXPERIENCE

Autonomous Husky for Trail Detection

Toronto Robotics and AI Laboratory (TRAIL) September 2023-June 2024

- Integrated and calibrated a 4K camera into ROS2 for semantic segmentation and LiDAR-based 3D perception.
- Trained an LEDNet CNN with a custom trail dataset, achieving 93.7% pixel accuracy for real-time trail segmentation.
- Optimized image processing to extract path centrelines and navigation targets for autonomous traversal.

Quadrotor Drone Racing Competition

Development of Unmanned Aerial Systems January 2024-April 2024

- Developed and implemented RRT* for 2D optimal trajectory planning, achieving second place in a competition by completing the track in 18.4 seconds.
- Utilized a simulation environment to debug planning algorithms, and tuned parameters to balance path curvature and velocity.
- Generated smooth 3D trajectories with 8-degree polynomial fits to ensure stable and efficient navigation through complex tracks.

Turtlebot Mapping, Navigation & Image Recognition

Mechatronics Systems Design & Integration January 2023-April 2023

- Implemented a reward-based RL algorithm to explore and map an unknown environment. Transitioned from rewarding open spaces to unknown terrain using the RGB-D Turtlebot camera.
- Utilized the move_base package to reach various waypoints and recognize the respective target image with an OpenCV pipeline.

Maze Navigating Rover

Mechatronics Principles September 2022-December 2022

- Prototyped an omni-wheeled rover with an onboard Pi and 2 Arduinos to retrieve and deliver a package in a maze.
- Utilized the lidar point cloud data with the ROS Hector SLAM, AMCL, and move_base packages to map, localize, and navigate.
- Built a Gazebo sim to train algorithm and tune parameters.